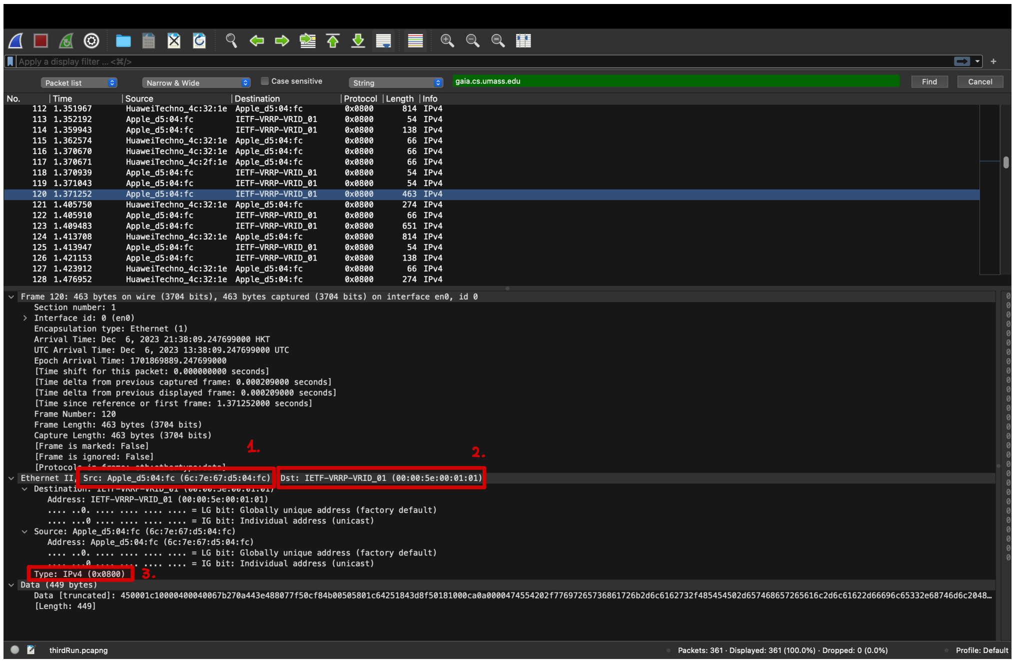
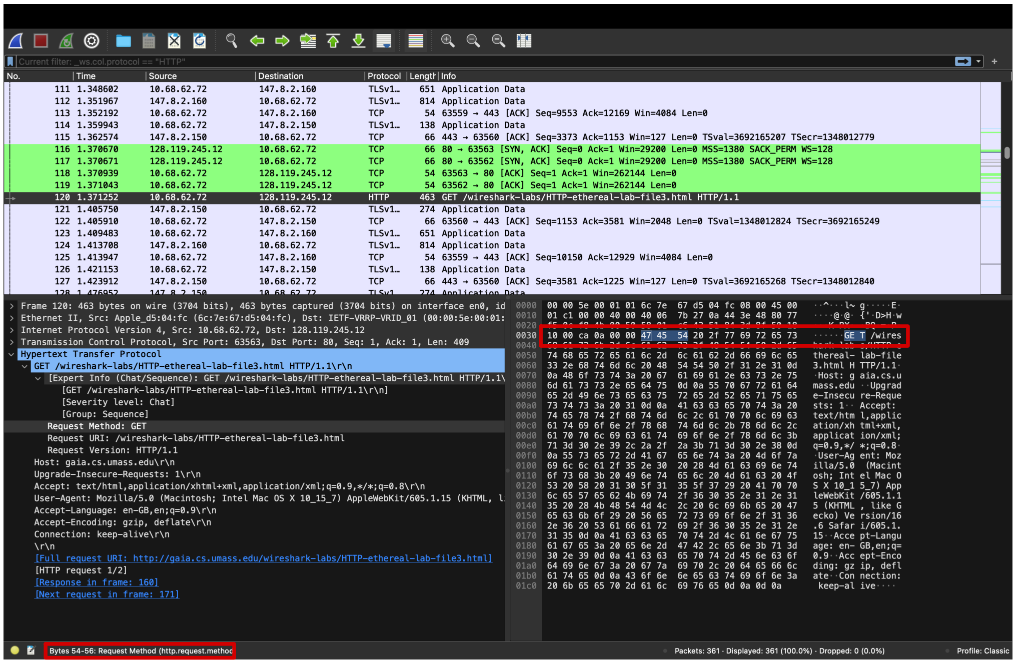
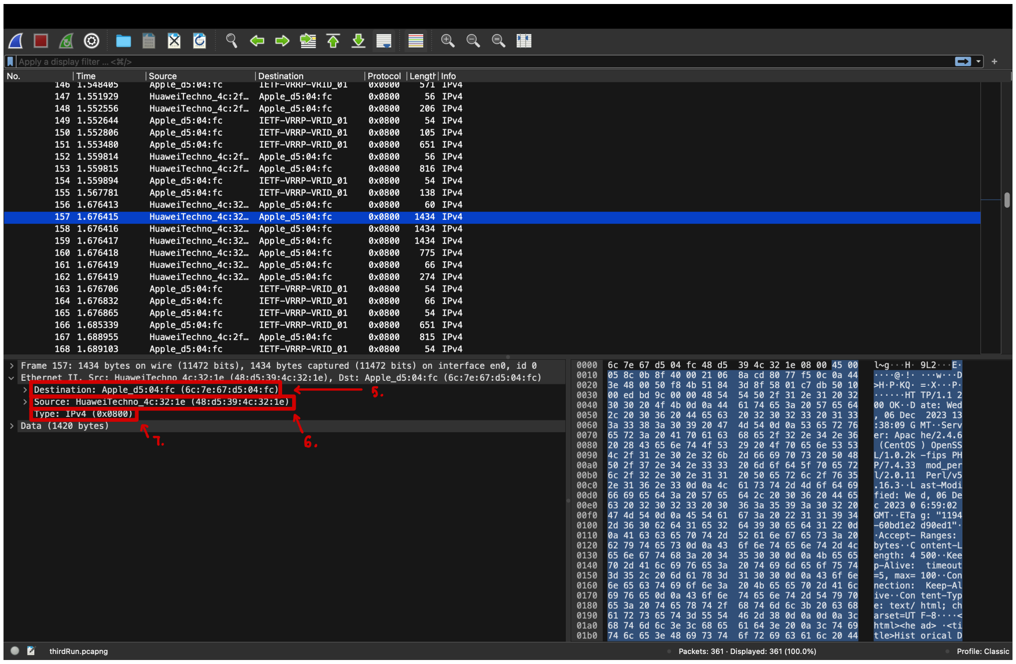
PART I



1. The 48-bit MAC address of the network adaptor on my computer is 6c:7e:67:d5:04:fc
2. The 48-bit destination MAC address in the Ethernet frame is 00:00:5e:00:01:01 This is not the MAC address of gaia.cs.umass.edu but the MAC address of the network adaptor of the router that my computer connected to.
3. The hexadecimal value for the two-byte type field in the Ethernet frame is 0x0800 and the upper layer protocol this corresponds to is IPv4.

4. There are 54 bytes from the start of The Ethernet frame the ASCII “G” in “GET” appear in the Ethernet frame.



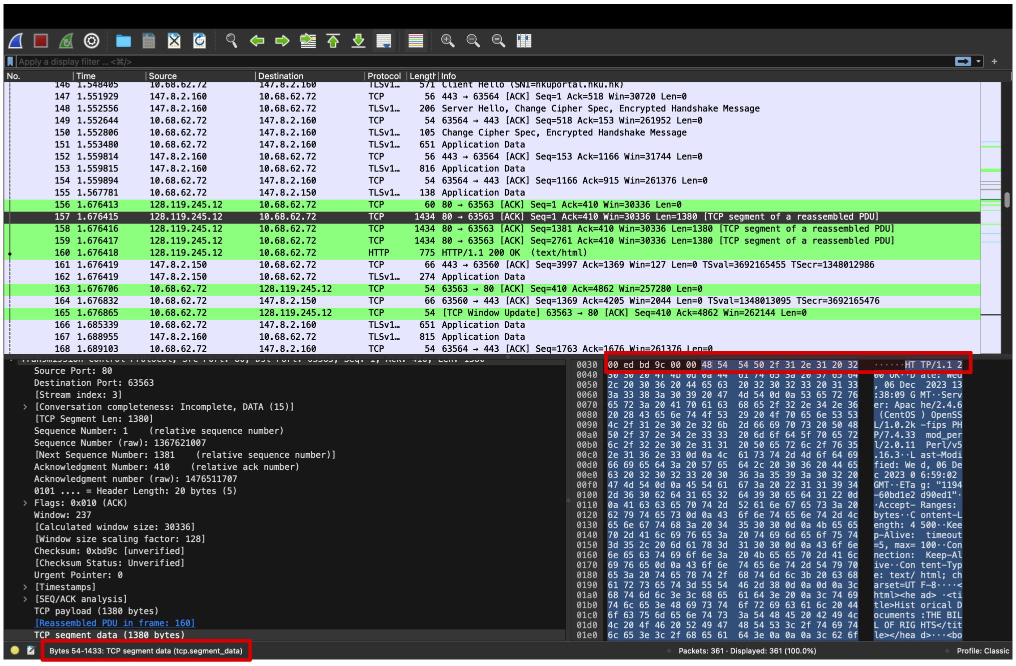


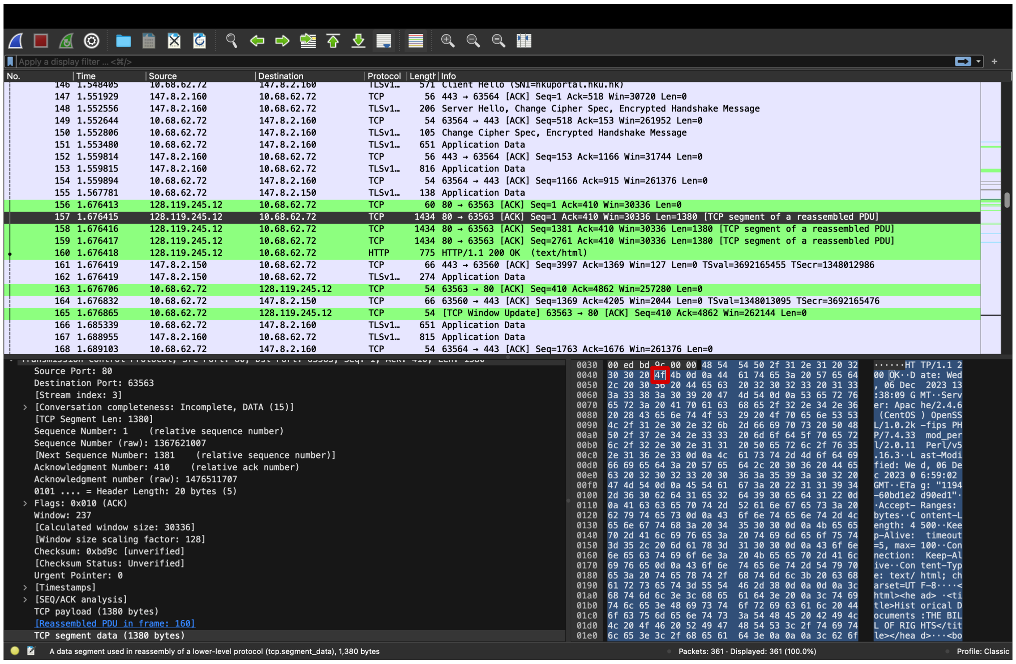
5. The source MAC address is 48:d5:39:4c:32:1e and this is the MAC address of the network adaptor of the router that my computer connected to.

6. The destination MAC address is 6c:7e:67:d5:04:fc and this is the MAC address of the network adaptor of my computer.

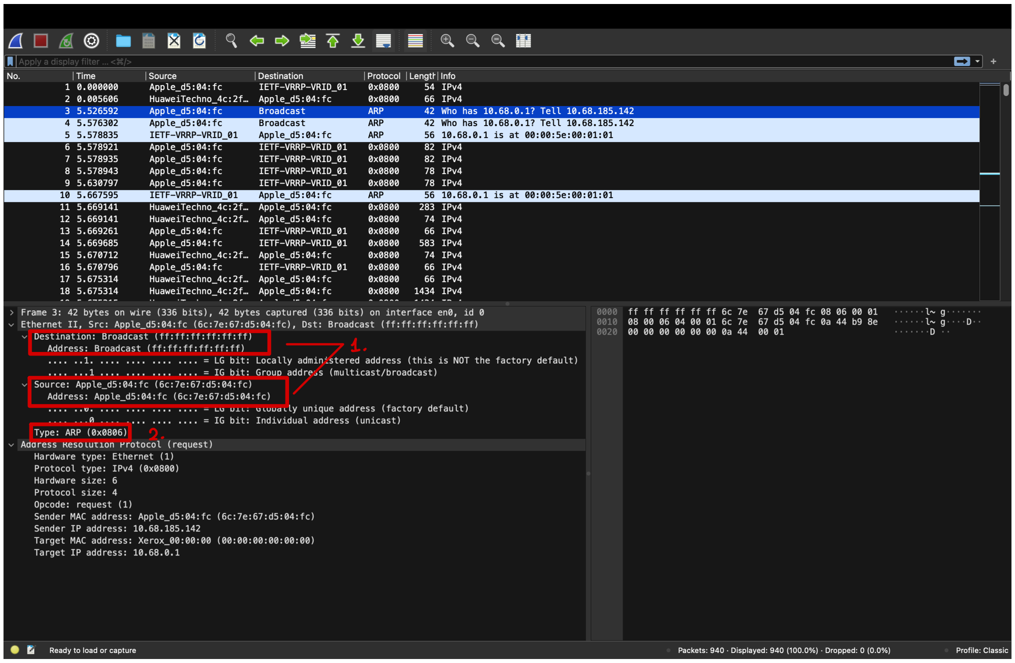
7. The hexadecimal value for the two-byte type field in the Ethernet frame is 0x0800 and the upper layer protocol this corresponds to is IPv4.

8. There are 54 bytes from the start of the Ethernet frame the ASCII “H” in “HTTP” appear in the Ethernet frame. So, here are 54 + 13 = 67 bytes from the start of the Ethernet frame the ASCII “O” in ”OK” appear in the Ethernet frame.





PART II



1. The hexadecimal values for the source and destination MAC addresses in the Ethernet frame containing this ARP request message are 6c:7e:67:d5:04:fc and ff:ff:ff:ff:ff:ff respectively.

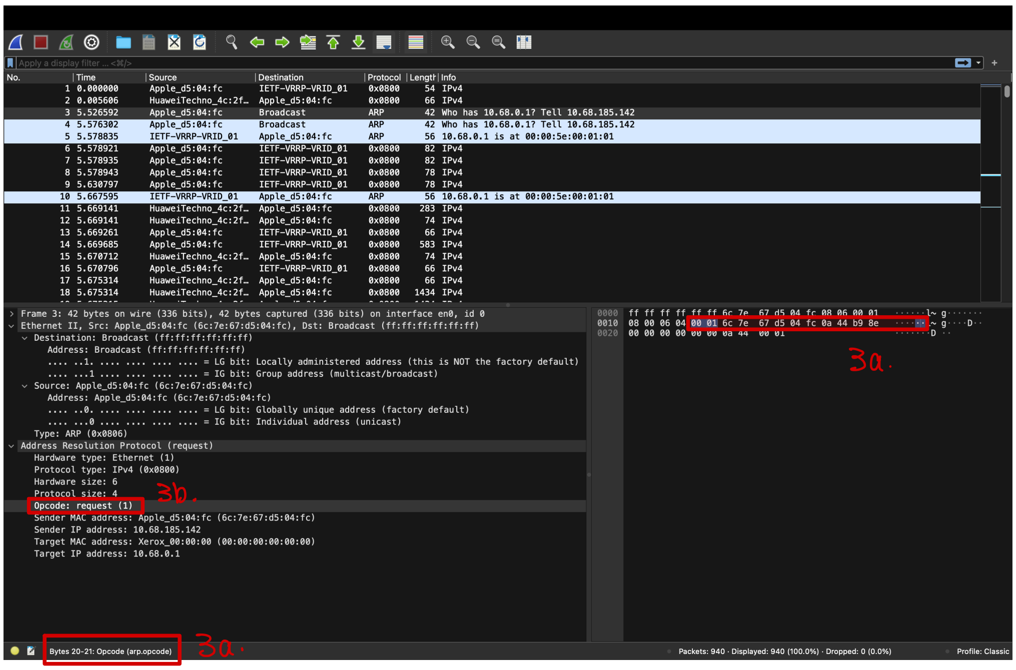
2. The hexadecimal value for the two-byte type field in the Ethernet frame is 0x0806 and the upper layer protocol this corresponds to is ARP protocol.

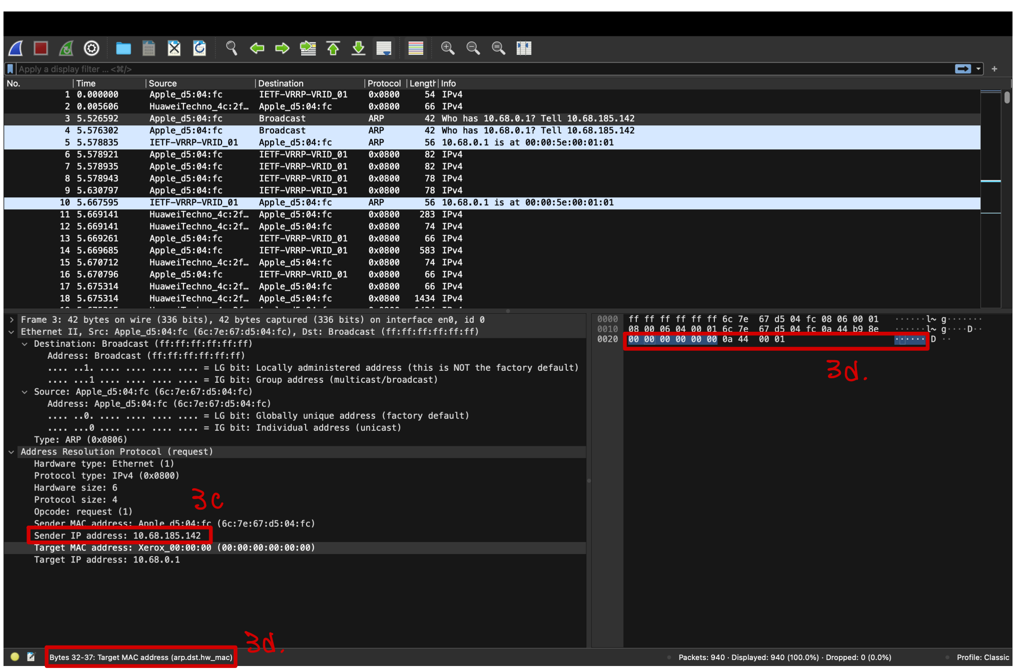
3. (a) The ARP opcode field begins from 20 bytes of the beginning of the Ethernet frame.

   (b) The value of the opcode field is 1.

   (c) The ARP message contains the IP address of the sender which is 10.68.185.142

   (d) The “question” in the ARP request message start from 32 bytes of the beginning of the Ethernet frame.

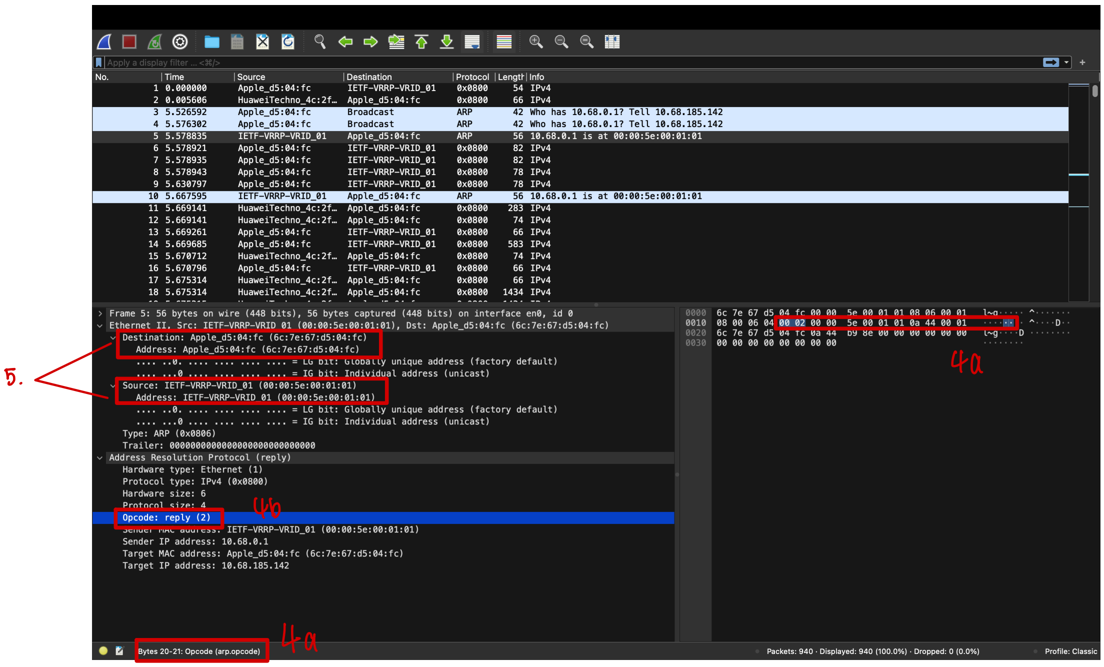


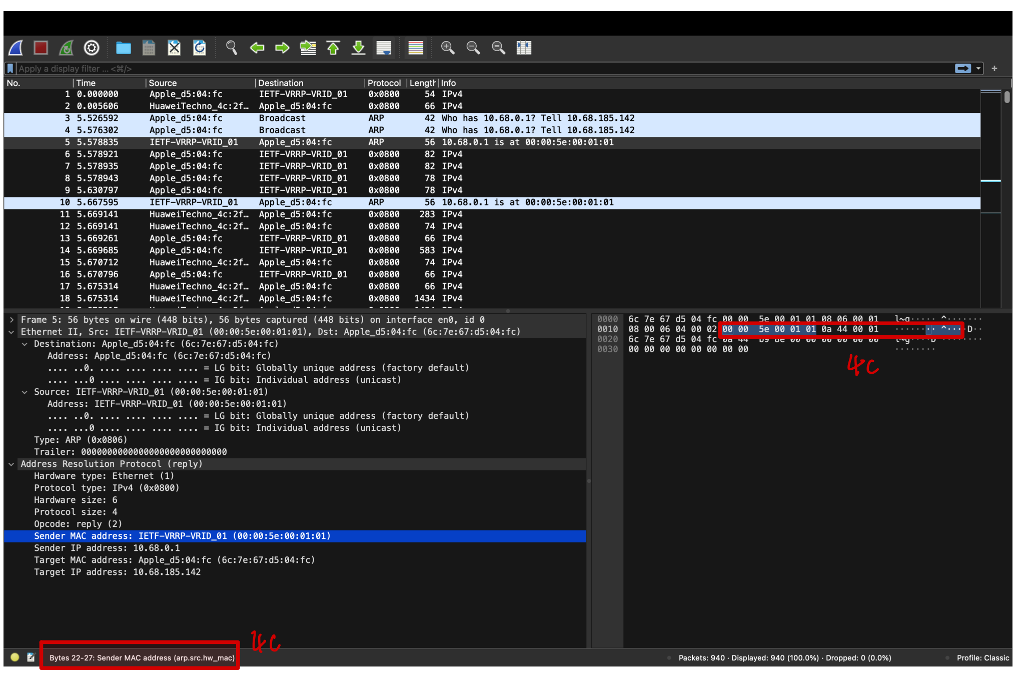


4. (a) The ARP opcode field begins from 20 bytes of the beginning of the Ethernet frame.

   (b) The value of the opcode field is 2.

   (c) The “answer” in the ARP response message start from 22 bytes of the beginning of the Ethernet frame.





5. The hexadecimal values for the source and destination MAC addresses in the Ethernet frame containing this ARP reply message are 00:00:5e:00:01:01 and 6c:7e:67:d5:04:fc respectively. (Please refer to the figure in the previous page).